

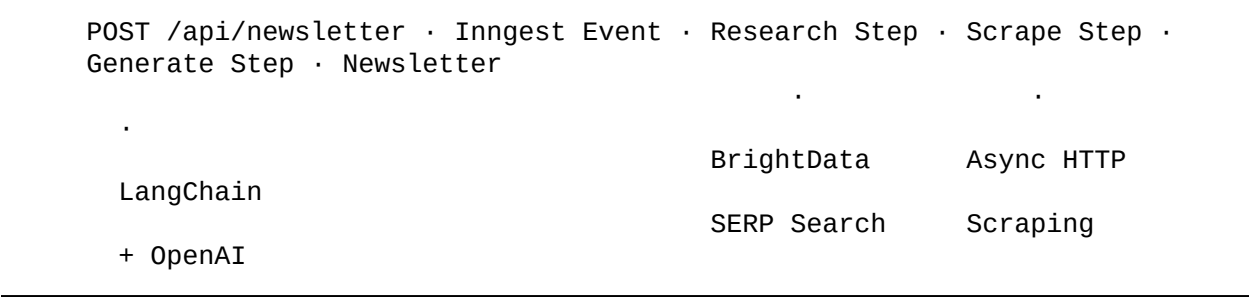
Scaleable Web AI Agent

A **scalable newsletter generation agent** built with **FastAPI** and **Inngest** for async workflow orchestration, using **Bright Data** for web research.

· Features

- **Newsletter Generation** · AI-powered newsletter creation from web research
- **SERP Search** · Bright Data integration for search engine results
- **Web Scraping** · Async content extraction from search results
- **Inngest Workflows** · Serverless, scalable function orchestration
- **LangChain** · LLM-powered content synthesis

.. Architecture



· File Structure

| File | Purpose | |-----|-----| | `main.py` | FastAPI + Inngest workflow (80 lines) | | `newsletter_service.py` | SERP search and async web scraping | | `custom_types.py` | NewsletterRequest Pydantic model |

· Getting Started

```
cd Scaleable-Web-AI-Agent
uv sync
uv run python main.py
```

Environment Variables (.env)

```
OPENAI_API_KEY=sk-...
BRIGHT_DATA_API_KEY=...
```

· Logic Flow

- 1 **Request** · Client sends newsletter topic via API
- 2 **SERP Search** · Bright Data fetches search results for the topic

- 3 **Web Scraping** · Top results scraped for content
 - 4 **Synthesis** · LangChain + OpenAI generates newsletter from scraped content
 - 5 **Response** · Generated newsletter returned
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· Dependencies

FastAPI, Inngest, langchain-brightdata, langchain, python-dotenv

· License

Educational project · use freely for learning and reference.